BASE CHANGE COMPATIBILITY OF THE ETA CONSTRUCTION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The maps (eta_f M)^i tensor B -> g^i N^i tensor g^{i+1} N^{i+1} and (eta_g N)^i -> g^i N^i tensor g^{i+1} N^{i+1} are inclusions of direct summands. invoked at proof lines 28120-28123. Lets the comparison be checked after localization.

Source reference. Original lemma 0F7W: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28050–28050. Cited at proof lines 28122-28123 for the direct summand inclusions on both sides; the conclusion is used inside the counted span.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ be a bounded complex of finite free A-modules. Assume f maps to a nonzerodivisor g in B and $I_i(M^\bullet, f)$ is a principal ideal for all $i \in \mathbf{Z}$. Then there is a canonical isomorphism $\eta_f M^\bullet \otimes_A B = \eta_g(M^\bullet \otimes_A B)$.*

Proof. Set $N^i = M^i \otimes_A B$. Observe that $f^i M^i \otimes_A B = g^i N^i$ as submodules of $(N^i)_g$. The maps

$$(\eta_f M)^i \otimes_A B \to g^i N^i \otimes g^{i+1} N^{i+1} \quad \text{and} \quad (\eta_g N)^i \to g^i N^i \otimes g^{i+1} N^{i+1}$$

are inclusions of direct summands by Lemma 0F7W. Since their images agree after localizing at g we conclude. □

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